

AUDITING AI-BASED ESG SCORES:

An Explainable, Fairness-Aware, and Greenwashing-Oriented Framework for ESG Governance

1. MOTIVATION

ESG scores increasingly influence investment decisions, corporate reputation, and sustainability reporting. Many AI-driven ESG scoring systems operate as black boxes, raising concerns regarding transparency, fairness, robustness, and accountability.

This study develops a comprehensive AI auditing framework to evaluate ESG scoring systems using explainability, robustness testing, fairness assessment, sustainability disclosure analysis, and greenwashing risk detection.

- ✓ Robustness of ESG predictions
- ✓ Explainability of ESG models
- ✓ Sector-specific bias
- ✓ Predictive power of sustainability disclosures
- ✓ Potential greenwashing risk

2. DATASETS

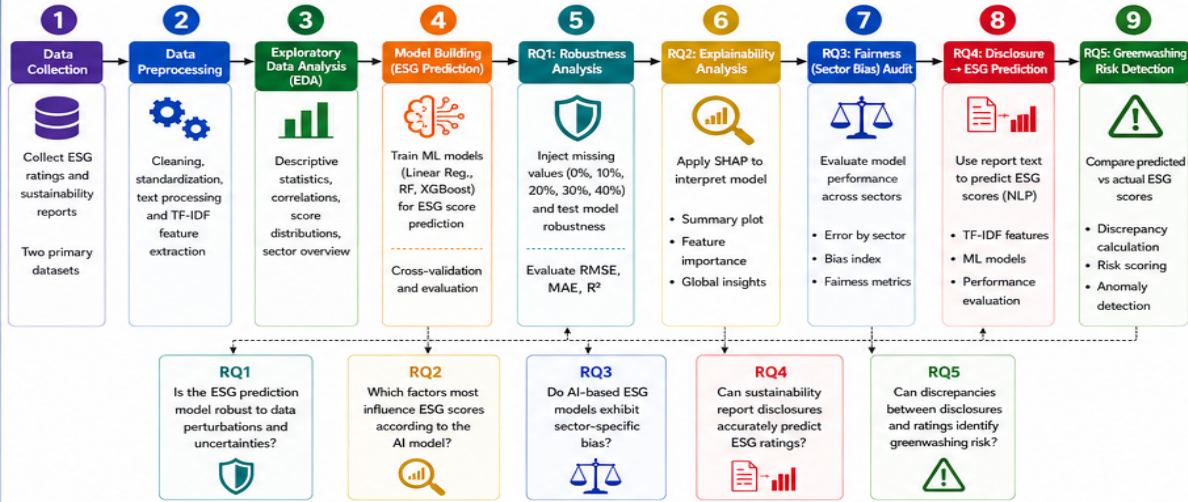
Dataset 1: Public Company ESG Ratings Dataset

- ESG Ratings
- Environmental Scores
- Social Scores
- Governance Scores
- Industry/Sector Information

Dataset 2: S&P 500 Sustainability Reports Dataset

- Sustainability Report Text
- ESG Scores
- Company Ticker
- Report Year

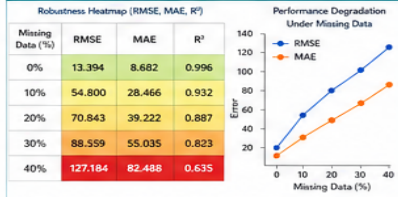
4. METHODOLOGY: STEP-BY-STEP WORKFLOW



3. RESEARCH QUESTIONS

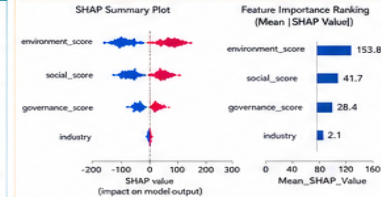
- RQ1** How robust are AI-generated ESG scores under missing or incomplete ESG information?
- RQ2** Which ESG factors most strongly influence AI-generated ESG scores?
- RQ3** Do AI-based ESG scoring systems exhibit sector-specific bias?
- RQ4** Can sustainability report disclosures predict ESG ratings?
- RQ5** Can discrepancies between disclosures and ESG ratings identify potential greenwashing risk?

RQ1 – Robustness Audit



As missing data increases, errors rise and R² drops significantly. The model is sensitive to severe ESG data incompleteness.

RQ2 – Explainability Audit



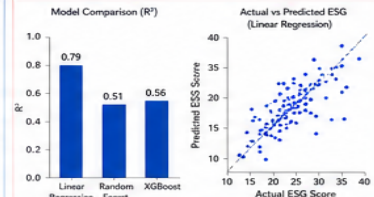
Environment score is the most influential driver of ESG predictions, followed by social and governance scores.

RQ3 – Fairness (Sector Bias) Audit



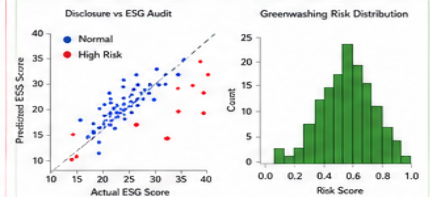
Significant sector-specific differences exist. Some sectors show much higher bias and error, indicating potential unfairness.

RQ4 – Disclosure -> ESG Prediction



Sustainability report text can predict ESG scores effectively. Linear Regression achieved the best performance (R² ≈ 0.79).

RQ5 – Greenwashing Detection



Discrepancy between predicted (disclosure) and actual ESG scores can reveal potential greenwashing risk.

6. CONTRIBUTIONS

- ✓ Developed an end-to-end AI auditing framework for ESG scoring.
- ✓ Integrated robustness, explainability, fairness, and greenwashing detection.
- ✓ Demonstrated that sustainability disclosures contain predictive ESG signals.
- ✓ Identified sector-specific biases in AI-based ESG scoring models.
- ✓ Proposed a disclosure-score discrepancy approach for greenwashing risk detection.
- ✓ Provided actionable insights for responsible AI governance in ESG.

7. CONCLUSION

This study presents a comprehensive AI auditing framework to examine ESG scoring models from multiple governance perspectives. Our results show that models are vulnerable to missing data, heavily influenced by environmental factors, and exhibit sector-specific biases. Sustainability reports can predict ESG ratings with reasonable accuracy, and disclosure-score discrepancies can indicate potential greenwashing risk. This framework supports more transparent, accountable, and trustworthy AI-driven ESG governance and can aid regulators, investors, and organizations in responsible decision-making.

8. IMPLICATIONS & FUTURE WORK

- Implications**
- Enhances transparency and accountability
 - Supports fair AI systems
 - Strengthens investor trust
 - Promotes sustainable capital markets

- Future Work**
- Incorporate external news and media data
 - Use advanced NLP (Transformers)
 - Temporal analysis of ESG trends
 - Causal fairness methods

- Impact**
- Contributes to the responsible governance of intelligent systems for a more sustainable and equitable future.